Microcontrollers			
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TITLE: Waveform Generation using DAC using V-Lab.

AIM: To write simple programs in C

i) To generate waveforms using DAC.

ii) To understand the use of addressing modes of 8051 while programming.

THEORY:

The digital-to-analog converter (DAC) is a device widely used to convert digital pulses to analog signals. There are two methods of creating a DAC: binary weighted and R/2R ladder. The vast majority of integrated circuit DACs, including the MC1408 (DAC0808) uses the R/2R method since it can achieve a much higher degree of precision. The first criterion for judging a DAC is its resolution, which is a function of the number of binary inputs. The common ones are 8, 10, and 12 bits. The number of data bit inputs decides the resolution of the DAC since the number of analog output levels is equal to 2″, where *n* is the number of data bit inputs. Therefore, an 8-input DAC such as the DAC0808 provides 256 discrete voltage (or current) levels of output. Similarly, the 12-bit DAC provides 4096 discrete voltage levels. There are also 16-bit DACs, but they are more expensive.

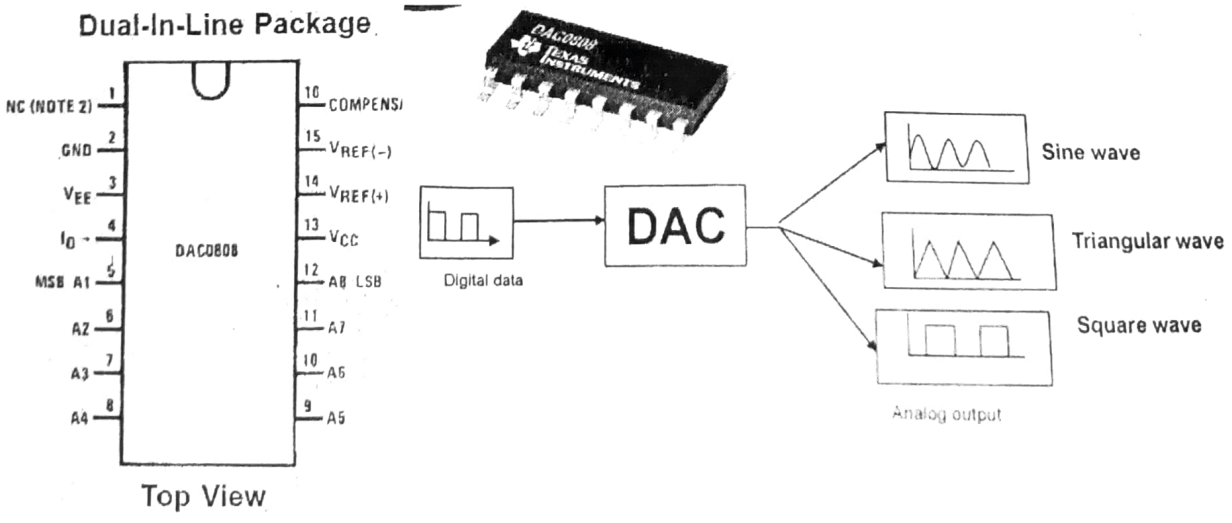


Figure: DIP Package of DAC 0808

MC1408 DAC (or DAC0808)

In the MC1408 (DAC0808), the digital inputs are converted to current (I_{out}), and by connecting a resistor to the I_{out} pin, we convert the result to voltage. The total current provided by the I_{out} pin is a function of the binary numbers at the DO – D7 inputs of the DAC0808 and the reference current (I_{ref}), and is as follows:

$$I_{out} = I_{ref}\left(\frac{D7}{2} + \frac{D6}{4} + \frac{D5}{8} + \frac{D4}{16} + \frac{D3}{32} + \frac{D2}{64} + \frac{D1}{128} + \frac{D0}{256}\right)$$

where DO is the LSB, D7 is the MSB for the inputs, and I_{ref} is the input current that must be applied to pin 14. The I_{ref} current is generally set to 2.0 mA. Figure 13-18 shows the generation of current reference (setting $I_{ref} = 2$ mA) by using the standard 5-V power supply and IK and 1.5K-ohm standard resistors. Some DACs also use the zener diode (LM336), which overcomes any fluctuation associated.

INTERFACING

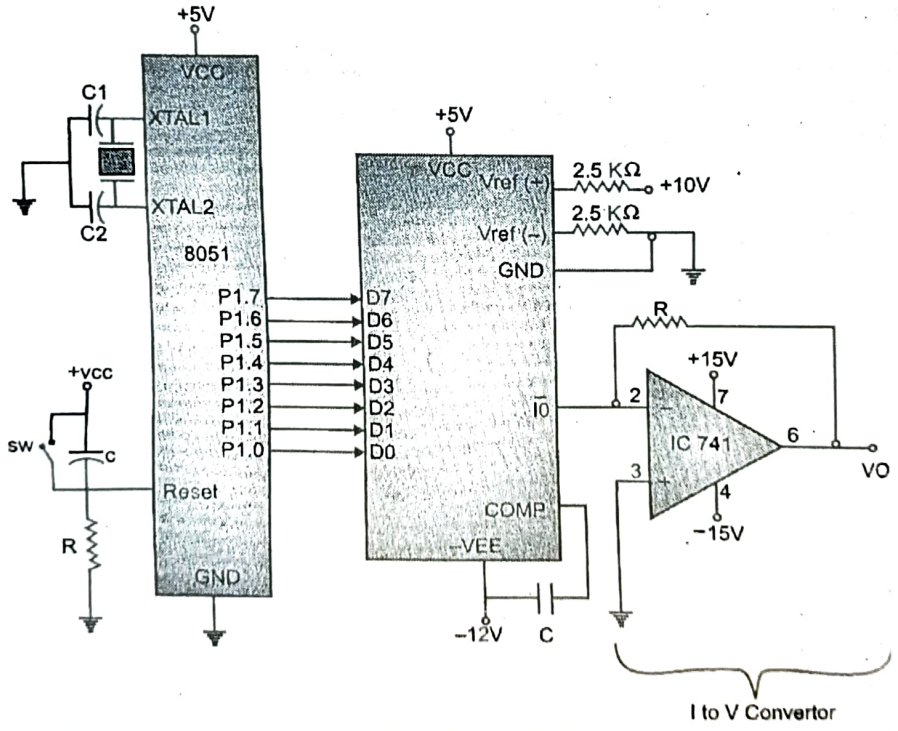


Figure: Interfacing diagram of DAC with 8051 for waveform generation.

We use 8-bit DAC 0808 which converts digital data into equivalent analog current. Hence, we require an I to V converter to convert this current into equivalent voltage.

So, V0= Iout * R.

Example:

1. If data = 00H i.e. [00000000], Vref= 10 V

   $$Vo=10(\frac{0}{2}+\frac{0}{4}+\frac{0}{8}+\frac{0}{16}+\frac{0}{32}+\frac{0}{64}+\frac{0}{128}+\frac{0}{256})$$

   i.e. V0= 0Volts

2. If data = 80H i.e. [10000000], Vref= 10 V

   $$Vo=10(\frac{1}{2}+\frac{0}{4}+\frac{0}{8}+\frac{0}{16}+\frac{0}{32}+\frac{0}{64}+\frac{0}{128}+\frac{0}{256})$$

   i.e. Vo= 5 Volts

3. If data = FFH i.e. [11111111], Vref= 10 V

   $$Vo=10(\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\frac{1}{16}+\frac{1}{32}+\frac{1}{64}+\frac{1}{128}+\frac{1}{256})$$

   i.e. Vo= 10 Volts

PROGRAM:

CONCLUSION:

QUESTIONS:

1. Write a code and generate a sawtooth waveform using DAC.
2. Write a code to generate a sine waveform using DAC.

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